AAE 416 / AAE 590: Viscous Flow

Fall 2025
Credit: 3

Course Information

- **Instructor:** Tom I-P. Shih (Professor of Aeronautics and Astronautics)
- **Teaching Assistant:** Dylan Hsieh
- **Pre-requisite:** any first course in fluid mechanics in aerospace, mechanical, chemical, or civil engineering (e.g., AAE 333 if from Purdue's School of Aeronautics and Astronautics)

Course Description

This course provides students with fundamentals in viscous flows of Newtonian fluids. The course starts by examining assumptions invoked in theoretical, experimental, and computational studies of flow problems, where viscosity, thermal conductivity, and species diffusivity cannot be neglected. Next, equations governing incompressible and compressible flows with and without chemical reactions are derived. This is followed by using differential and integral methods to obtain exact and approximate solutions for parallel, nearly parallel, and creeping flows. Afterwards, an introduction is given on linear theory of hydrodynamic stability including transition to turbulence. The focus throughout the course is on flow mechanisms quantified by exact and approximate solutions. Another focus is on how solutions obtained on simple configurations could be used as reduced-order models in assisting the study of more complicated configurations by anticipating flow structures that could take place and in estimating the length and time scales needed to resolve those flow structures in experimental and computational studies.

Course Material

- **Textbook (optional):** Viscous Fluid Flow (3rd edition) by White
- **Brightspace:** syllabus, lecture notes, recorded lectures, tutorials, homework assignments and solutions, and handouts with summaries on special topics

Course Outline

- **Introduction:** Review of basic concepts. Significance of viscous phenomena such as secondary flows and mechanisms that create them.
- **Governing Equations:** Derivation of continuity, species balance, Navier-Stokes, and total energy equations for reacting and nonreacting flows that include coupling effects. Boundary-layer concepts and equations. Mathematical character of PDEs used to model fluid flow and appropriate initial and boundary conditions.
- **Exact Solutions of Incompressible Laminar Flow:** Parallel flows. Low Reynolds number flows. Boundary-Layer flows (Blasius and Falkner-Skan solutions).
- **Integral Solutions of Laminar Boundary Layers:** Boundary-layer integral equations. Karman-Pohlhausen technique. Thwaite's method. Natural convection on a vertical plate.
- **Exact Solutions of Compressible Laminar Flow:** Compressible Couette flow. Compressible Boundary Layers with viscous dissipation on adiabatic and isothermal walls for subsonic and supersonic flows.
- **Introduction to Shock-Wave/Boundary-Layer Interactions.**
- **Introduction to Hydrodynamics Stability:** gravitational instability, inertial instability, instability due to surface tension, instability due to vorticity distribution in an inviscid fluid (Rayleigh, Fjortoft, Tollmien), Helmholtz instability.
- **Introduction to Turbulent Boundary Layers:** An introduction to turbulent boundary layers including length scales from integral to Kolmogorov, the law of the wall, and the law of the wake.

Learning Objectives

- Be able to make meaningful assumptions and understand consequences of assumptions when solving engineering problems involving fluid flow.

- Be able to formulate viscous flow problems, including boundary conditions.
- Be able to explain mechanisms that lead to flow physics such as secondary flows, flow separation, and transition to turbulence.
- Be able to use exact and approximate solutions of simple viscous flow problems to do back-of-the-envelope estimates of more complicated viscous flow problems.

AAE 416 vs AAE 590

- For students in AAE 416, the homework will focus on applying theory and solution procedures to practical problems.
- **For students in AAE 590, one additional homework problem will be added to each homework set that require a deeper understanding on the derivation of equations and solution procedures.**
- Exams will be the same for both AAE 416 and AAE 590 students and will focus on applying theory and solution procedures to practical problems.

Course Grade

- HW: 70%
- Exam 1: 15%
- Exam 2: 15%
- No final exam.

Grading Scale

A: 90-100, A-: 85-89, B+: 80-84, B: 75-79, B-: 70-74, C+: 65-69, C: 60-64, C-:55-59, D+: 50-54, D: 45-49, D-: 40-44, F: <40